

JT808 Data Sending Protocols

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Version History

Version	Date	Modifier	Changes
1.0.0	2026-04-22	Zhuojie Cai	Initial version
1.0.1	2026-05-07	Zhuojie Cai	Removed deprecated information collection messages (0x8700/0x0700/0x8701/0x0701/0x8702/0x0702/0x0704/0x0705) from 4.7 and deprecated messages from 4.3, updated table of contents, section numbering, and terminology
1.0.2	2026-06-11	Zhuojie Cai	Restored 0x0704 batch location data upload (uplink) protocol

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1 Protocol Overview

JT/T 808 is the communication protocol and data format standard for **in-vehicle terminals and monitoring platforms** in the road transport vehicle satellite positioning system. **JT/T 808-2011/2013/2019** are its main versions. It uses **TCP/UDP** communication, with the platform as the **server** and the terminal as the **client**. The terminal actively connects to the platform (**persistent connection**), with messages exchanged bidirectionally at any time.

Extended Protocol Notes: This standard protocol (JT/T 808) only contains basic positioning and communication functions. For the following extended functions, please refer to the corresponding protocol documentation:

Extended Protocol	Description
Proactive Safety - Additional Information Items Protocol	Driving assistance, driver behavior monitoring, etc.
Additional Information Items Protocol	Location additional information item extensions (0x0200 extended area, customer-defined data)

Note: For other extended protocols or if you cannot find the required protocol definition, please contact **the software technical support team**.

Message Direction Rules:

- Uplink messages (Terminal to Platform): Message IDs start with **0x0**, such as 0x0001, 0x0200
- Downlink messages (Platform to Terminal): Message IDs start with **0x8**, such as 0x8001, 0x8103

Protocol Version Identification Rules:

JT/T 808 has three main versions: **2011**, **2013**, and **2019**, distinguished by the following methods:

Condition	Protocol Version	Message Header Length	Terminal Phone Number	Description
Province ID=0 and bit14=0	2011 Version	12 bytes	BCD[6] , 12 digits	Earliest version
Province ID≠0 and bit14=0	2013 Version	12 bytes	BCD[6] , 12 digits	bit14 is a reserved bit, default is 0
bit14=1	2019 Version	17 bytes	BCD[10] , 20 digits	bit14 is fixed at 1, added 1 byte protocol version number field

Parsing Steps:

1. Read message body property (offset 2, 2 bytes WORD)

2. Check **bit14** (i.e., **bit6** of the first byte of message body property): if value is 1 → 2019 version
3. When bit14=0, further distinguish by **Province ID** in the terminal registration message (0x0100):
 - Province ID=0 → 2011 version
 - Province ID≠0 → 2013 version
4. The 2019 version message header additionally contains **1 byte protocol version number** after the message body property, and the terminal phone number expands from **BCD[6]** to **BCD[10]**

Terminology

Term	Full Name/Meaning	Description
T	Terminal	In-vehicle terminal device, TCP/UDP communication client. In message flow, T→P indicates a message sent from terminal to platform
P	Platform	Monitoring platform server, TCP/UDP communication server. In message flow, P→T indicates a message sent from platform to terminal
Uplink	Uplink, Terminal → Platform	Message direction: terminal sends to monitoring platform. In JT/T 808, message IDs starting with <code>0x0</code> (such as <code>0x0001</code> , <code>0x0200</code>) indicate uplink messages

Downlink	Downlink, Platform → Terminal	Message direction: monitoring platform sends to terminal. In JT/T 808, message IDs starting with 0x8 (such as 0x8001 , 0x8103) indicate downlink messages
TLV	Type-Length-Value	A common data encoding format consisting of three parts: Tag (ID), Length, and Value. Additional information items in this protocol use this structure
BCD	Binary-Coded Decimal	8421 BCD code, an encoding method that uses 4 binary bits to represent one decimal digit. This protocol uses this encoding for phone numbers, time, and other fields
Packet Splitting	Packet Splitting / Fragmentation	When a single message is too long, it is split into multiple data packets for transmission. Split packets are identified by message package encapsulation items (total package count + package sequence number)
Check Code	Check Code / XOR Checksum	1-byte XOR checksum value, calculated by XORing byte-by-byte from the first byte of the message header to the byte before the check code, used to determine if the message is complete

Escape	Escape	When the message content contains bytes identical to the flag byte (<code>0x7E</code>) or escape character (<code>0x7D</code>), they are converted to escape sequences: <code>0x7E</code> → <code>0x7D 0x02</code> , <code>0x7D</code> → <code>0x7D 0x01</code>
Flag Byte	Flag Byte	Fixed value <code>0x7E</code> , used to mark the start and end of a JT/T 808 message. The receiver uses the flag byte to identify the beginning and end of a complete message
GPS	Global Positioning System	Global Positioning System, the U.S. satellite navigation system. In this protocol, GPS positioning refers to position data obtained through the GPS module
GNSS	Global Navigation Satellite System	Global Navigation Satellite System, a general term covering GPS, BeiDou, GLONASS, Galileo, and all other satellite navigation systems
BeiDou	BeiDou / BDS	China's independently developed satellite navigation system. In this protocol, "BeiDou positioning" refers to position data obtained through the BeiDou module
VSS	Vehicle Speed Sensor	Vehicle speed sensor, used to detect vehicle traveling speed

ACC	Accessory	Accessory power state, a state bit in the vehicle electrical system. ACC on means the vehicle key is in the accessory/ON position, ACC off means ignition off/LOCK position
IC Card	Integrated Circuit Card	In this protocol, refers to the road transport certificate IC card, used to store and read driver identity information
CAN Bus	Controller Area Network Bus	Controller Area Network bus, used for real-time data communication between in-vehicle devices
VIN	Vehicle Identification Number	Vehicle Identification Number, a 17-digit unique code used to identify a vehicle
RSA	Rivest–Shamir–Adleman	An asymmetric encryption algorithm. In this protocol, the terminal and platform each hold a pair of RSA public/private keys for encrypted data transmission
IMEI	International Mobile Equipment Identity	International Mobile Equipment Identity, a 15-digit unique number identifying a mobile phone. In the 2019 version protocol, the terminal authentication message carries the IMEI

ICCID	Integrated Circuit Card Identity	Integrated Circuit Card Identity, the unique identifier of a SIM card. In the 2019 version protocol, the terminal attribute response carries the ICCID
Register	Register	When a terminal first connects to the platform, it reports vehicle and terminal information, and the platform returns an authentication code
Authenticate	Authenticate	The process where the terminal uses the authentication code (or in the 2019 version, additionally carries IMEI and software version number) to prove its identity legitimacy to the platform
Heartbeat	Heartbeat	An empty message periodically sent by the terminal to maintain the TCP connection and prove the terminal is still online
Transparent Transmission	Transparent Transmission	Data is not parsed or processed, but forwarded directly. In this protocol, <code>0x0900</code> (uplink transparent transmission) and <code>0x8900</code> (downlink transparent transmission) are used to transmit third-party protocol data

Additional Information Item	Additional Information Item	Optional extended fields in location information reporting (<code>0x0200</code>), using TLV structure, can carry mileage, fuel level, CAN data, etc.
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1.1 Device Registration and Authentication Flow

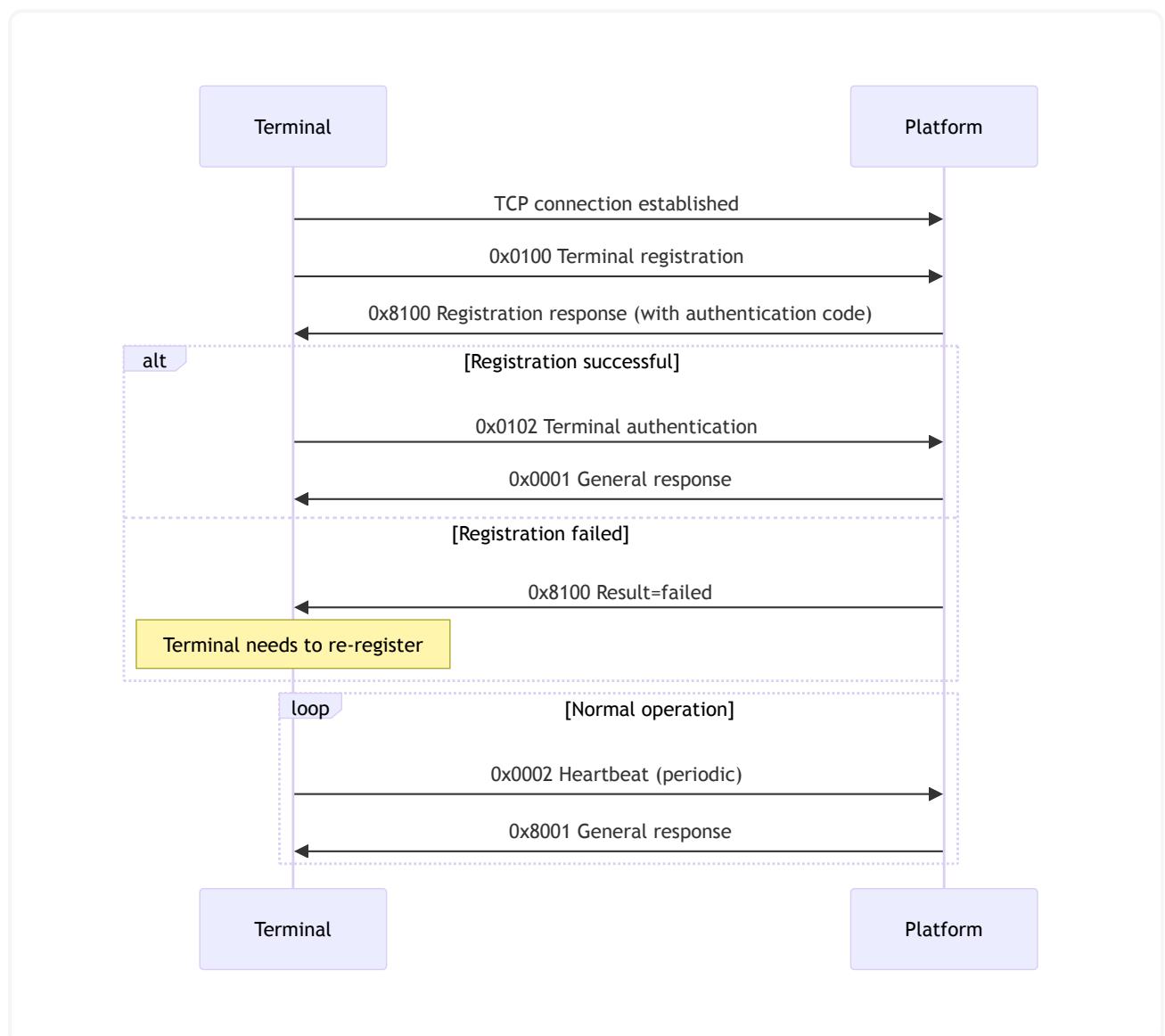
After the terminal connects to the platform, registration and authentication must be completed for normal operation.

Standard Registration and Authentication Flow

Step	Message ID	Direction	Description
1	—	T→P	TCP connection established
2	0x0100	T→P	Terminal registration (province ID, city/county ID, manufacturer ID, terminal model, terminal ID, license plate, etc.)
3	0x8100	P→T	Registration response, containing result and authentication code

4	0x0102	T→P	Terminal authentication (carrying authentication code, 2019 version adds IMEI and software version number)
5	0x0001	P→T	General response
6	0x0002	T→P	Heartbeat (periodic, connection maintenance)

Sequence Diagram



Reconnection Authentication Flow

Step	Message ID	Direction	Description
1	—	T→P	TCP connection established
2	0x0102	T→P	Terminal authentication (using saved authentication code)
3	0x0001	P→T	General response (continue working if successful)

2 Protocol Basics

2.1 Data Types

Type	Description
BYTE	Unsigned single-byte integer (8 bits)
WORD	Unsigned double-byte integer (16 bits)
DWORD	Unsigned four-byte integer (32 bits)
BCD[n]	8421 BCD encoding, n bytes
STRING	GBK encoded string
BYTE[n]	n bytes of data

Transmission Rules: Big-endian mode, high byte transmitted first.

2.2 Message Structure

Part	Description
Flag Byte	Fixed 0x7E
Message Header	See message header structure below
Message Body	Customized for each message
Check Code	XOR checksum, 1 byte
Flag Byte	Fixed 0x7E

Escape Rules: Escape processing must be strictly executed in order

- Step 1:** Escape **0x7D** in the message content (check code, message header, message body) → **0x7D 0x01**
- Step 2:** Escape **0x7E** in the result of step 1 → **0x7D 0x02**

Note: Process 0x7D first, then 0x7E. The order must not be reversed.

Example: Raw data 0x30 0x7E 0x08 0x7D 0x55 → After escaping 0x7E 0x30 0x7D 0x02 0x08 0x7D 0x01 0x55 0x7E

2.3 Message Header Structure

Offset	Field	Type	2011	2013	2019
0	Message ID	WORD	—	—	—
2	Message Body Property	WORD	See format below	See format below	See format below
4	Protocol Version Number	BYTE	No such field	No such field	[2019] Added
5	Terminal Phone Number	BCD[6]	Pad with 0 if insufficient	Pad with 0 if insufficient	[2019] BCD[10], offset 5
10	Message Serial Number	WORD	Cyclic increment from 0	Cyclic increment from 0	Cyclic increment from 0

12	Message Package Encapsulation	—	Present when split	Present when split	Present when split
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Note: The 2011 version and 2013 version have the same message header structure, neither has the protocol version number field.

Message Body Property Format (16 bits):

Bit	2013	2019
bit0~bit9	Message body length	Message body length
bit10~bit12	Data encryption method	Data encryption method
bit13	Subpackage flag	Subpackage flag
bit14	Reserved	Version identifier (fixed at 1)
bit15	Reserved	Reserved

Message Package Encapsulation (when split):

Field	Type	Description
Total Package Count	WORD	Total packages after splitting
Package Sequence Number	WORD	Starting from 1

Check Code: From the first byte of the message header to the byte before the check code, XOR byte by byte.

2.3.1 2013 Version Message Header Parsing

Example Message (0x0001 Terminal General Response):

7E 00 01 00 05 01 38 12 34 56 78 00 01 00 01 80 01 00 B4 7E

Message header is 12 bytes total, structure as follows:

Offset	Bytes	Field Name	Example Value	Description
0	2	Message ID	00 01	0x0001 Terminal general response
2	2	Message Body Property	00 05	Length=5, no encryption, not split
4	6	Terminal Phone Number	01 38 12 34 56 78	BCD[6], 12 digits, pad with 0 if insufficient
10	2	Message Serial Number	00 01	First message

Message Body: 00 01 80 01 00 (Response serial number + Response ID + Result)

Check Code: B4

2.3.2 2019 Version Message Header Parsing

Example Message (0x0001 Terminal General Response):

7E 00 01 40 05 01 00 00 00 00 01 38 12 34 56 78 00 00 01 00 01 80 01 00 F5 7E

Message header is 17 bytes total, structure as follows:

Offset	Bytes	Field Name	Example Value	Description
0	2	Message ID	00 01	0x0001 Terminal general response
2	2	Message Body Property	40 05	Length=5, no encryption, not split, Version identifier=1

4	1	Protocol Version Number	01	[2019 Added]
5	10	Terminal Phone Number	00 00 00 00 01 38 12 34 56 78 00	BCD[10], 20 digits, pad with 0 if insufficient
15	2	Message Serial Number	00 01	First message

Message Body: 00 01 80 01 00 (Response serial number + Response ID + Result)

Check Code: F5

3 Message Overview

3.1 Uplink Messages (Terminal to Platform)

Message ID	Name	Category	Notes
0x0001	Terminal general response	Terminal management	
0x0002	Terminal heartbeat	Terminal management	Message body is empty
0x0003	Terminal logout	Terminal management	Message body is empty
0x0004	Query server time request	Terminal management	[2019] Message body is empty
0x0005	Terminal resend package request	Terminal management	[2019]
0x0100	Terminal registration	Terminal management	

0x0102	Terminal authentication	Terminal management	[2019] Adds IMEI and version number
0x0104	Query terminal parameters response	Terminal management	
0x0107	Query terminal attributes response	Terminal management	
0x0108	Terminal upgrade result notification	Terminal management	
0x0200	Location information reporting	Location and alarm	Core message
0x0201	Location information query response	Location and alarm	
0x0500	Vehicle control response	Vehicle control	
0x0608	Query area or route data response	Vehicle management	[2019]
0x0800	Multimedia event information upload	Multimedia	
0x0801	Multimedia data upload	Multimedia	
0x0802	Stored multimedia data search response	Multimedia	
0x0805	Camera immediate capture command response	Multimedia	
0x0900	Uplink transparent transmission	Data transmission	

0x0901	Data compression report	Data transmission	
0x0A00	Terminal RSA public key	Encryption	

3.2 Downlink Messages (Platform to Terminal)

Message ID	Name	Category	Notes
0x8001	Platform general response	Terminal management	
0x8003	Resend package request	Terminal management	
0x8004	Query server time response	Terminal management	[2019]
0x8100	Terminal registration response	Terminal management	
0x8103	Set terminal parameters	Terminal management	
0x8104	Query terminal parameters	Terminal management	Message body is empty
0x8105	Terminal control	Terminal management	
0x8106	Query specified terminal parameters	Terminal management	[2019]
0x8107	Query terminal attributes	Terminal management	Message body is empty
0x8108	Deploy terminal upgrade package	Terminal management	
0x8201	Location information query	Location and alarm	Message body is empty

0x8202	Temporary location tracking control	Location and alarm	
0x0704	Batch location data upload (uplink)	Location and alarm	
0x8203	Manual alarm confirmation	Location and alarm	[2019]
0x8204	Link detection	Location and alarm	[2019] Message body is empty
0x8300	Text information下发	Information	
0x8400	Telephone callback	Telephone	
0x8401	Set telephone book	Telephone	
0x8500	Vehicle control	Vehicle control	
0x8600	Set circular area	Vehicle management	
0x8601	Delete circular area	Vehicle management	
0x8602	Set rectangular area	Vehicle management	
0x8603	Delete rectangular area	Vehicle management	
0x8604	Set polygon area	Vehicle management	
0x8605	Delete polygon area	Vehicle management	
0x8606	Set route	Vehicle management	
0x8607	Delete route	Vehicle management	

0x8608	Query area or route data	Vehicle management	[2019]
0x8800	Multimedia data upload response	Multimedia	
0x8801	Camera immediate capture command	Multimedia	
0x8802	Stored multimedia data search	Multimedia	
0x8803	Stored multimedia data upload command	Multimedia	
0x8804	Recording start command	Multimedia	
0x8805	Single stored multimedia data search upload command	Multimedia	
0x8900	Downlink transparent transmission	Data transmission	
0x8A00	Platform RSA public key	Encryption	

4 Message Details

4.1 Terminal Management

0x0001 Terminal General Response (Uplink)

Offset	Field	Type	Description
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0	Response serial number	WORD	Corresponds to the platform message serial number
2	Response ID	WORD	Corresponds to the platform message ID
4	Result	BYTE	0: Success/Ack 1: Failure 2: Message error 3: Not supported

2013 Version Example:7E 00 01 00 05 01 38 12 34 56 78 00 01 00 01 80 01 00 B4 7E

2019 Version Example:7E 00 01 40 05 01 00 00 00 00 01 38 12 34 56 78 00 01 00 01 80 01 00 F5 7E

0x8001 Platform General Response (Downlink)

Offset	Field	Type	Description
0	Response serial number	WORD	Corresponds to the terminal message serial number
2	Response ID	WORD	Corresponds to the terminal message ID
4	Result	BYTE	0: Success/Ack 1: Failure 2: Message error 3: Not supported 4: Alarm processing confirmed

2013 Version Example:7E 80 01 00 05 01 38 12 34 56 78 00 01 00 01 00 02 00 B7 7E

2019 Version Example:7E 80 01 40 05 01 00 00 00 00 01 38 12 34 56 78 00 01 00 01 00 02 00 F6 7E

0x0002 Terminal Heartbeat (Uplink)

Message body is empty.

2013 Version Example:7E 00 02 00 00 01 38 12 34 56 78 00 01 32 7E

2019 Version Example:7E 00 02 40 00 01 00 00 00 00 01 38 12 34 56 78 00 01 73 7E

0x0003 Terminal Logout (Uplink)

Message body is empty.

2013 Version Example:7E 00 03 00 00 01 38 12 34 56 78 00 01 33 7E

2019 Version Example:7E 00 03 40 00 01 00 00 00 00 01 38 12 34 56 78 00 01 72 7E

0x0004 Query Server Time Request (Uplink) [2019]

Message body is empty. Terminal queries the server for the current time.

2013 Version Example:

(N/A, this message does not exist in 2013 version)

2019 Version Example:7E 00 04 40 00 01 00 00 00 00 01 38 12 34 56 78 00 01 75 7E

0x8004 Query Server Time Response (Downlink) [2019]

Offset	Field	Type	Description
0	Server time	BCD[6]	UTC time, format YY-MM-DD-hh-mm-ss

2013 Version Example:

(N/A, this message does not exist in 2013 version)

2019 Version Example:7E 80 04 40 06 01 00 00 00 00 01 38 12 34 56 78 00 01 26 04 22 10 30 00 D3 7E

0x0005 Terminal Resend Package Request (Uplink) [2019]

Format is the same as 0x8003 Resend Package Request.

Offset	Field	Type	Description
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0	Original message serial number	WORD	Serial number of the first packet of the original message to be resent
4	Total resend packages	BYTE	n
5	Resend package ID list	BYTE[2*n]	Resend package sequence numbers in order

2013 Version Example:

(N/A, this message does not exist in 2013 version)

2019 Version Example: 7E 00 05 40 06 01 00 00 00 00 01 38 12 34 56 78 00 01 00 01 00 01 00 01 73 7E

0x8003 Resend Package Request (Downlink)

Offset	Field	Type	Description
0	Original message serial number	WORD	Serial number of the first packet of the original message to be resent
4	Total resend packages	BYTE	n
5	Resend package ID list	BYTE[2*n]	Resend package sequence numbers in order

2013 Version Example: 7E 80 03 00 06 01 38 12 34 56 78 00 01 00 01 00 01 00 01 B4 7E

2019 Version Example: 7E 80 03 40 06 01 00 00 00 00 01 38 12 34 56 78 00 01 00 01 00 01 00 01 F5 7E

0x0100 Terminal Registration (Uplink)

Offset	Field	Type	2011	2013	2019
0	Province ID	WORD	First two digits of GB/T 2260, 0 reserved	First two digits of GB/T 2260, 0 reserved	First two digits of GB/T 2260, 0 reserved
2	City/County ID	WORD	Last four digits of GB/T 2260, 0 reserved	Last four digits of GB/T 2260, 0 reserved	Last four digits of GB/T 2260, 0 reserved
4	Manufacturer ID	BYTE[5]	Manufacturer code	Manufacturer code	[2019] BYTE[11], composed of administrative division code + manufacturer ID
9	Terminal model	BYTE[8]	Pad with 0x00 if insufficient	BYTE[20], pad with 0x00 if insufficient	[2019] BYTE[30], start at 15
17	Terminal ID	BYTE[7]	Pad with 0x00 if insufficient	BYTE[7], start at 29, pad with 0x00 if insufficient	[2019] BYTE[30], start at 45
24	License plate color	BYTE	JT/T 415, 0 if not registered	JT/T 415, start at 36, 0 if not registered	[2019] JT/T 697.7-2014, start at 75

25	Vehicle identifier	STRING	VIN if license plate color is 0, otherwise license plate number	VIN if license plate color is 0, otherwise license plate number, start at 37	[2019] License plate, start at 76
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2013 Version Example: 7E 01 00 00 2D 01 38 12 34 56 78 00 01 00 23 00 64 42 53 4A 00 00 54 54 4D 30 30 31 00 00 00 00 00 00 00 00 00 00 00 00 31 32 33 34 35 36 37 01 BE A9 41 38 38 38 38 23 7E

[illegible]

0x8100 Terminal Registration Response (Downlink)

Offset	Field	Type	Description
0	Response serial number	WORD	Corresponds to the terminal registration message serial number
2	Result	BYTE	0: Success 1: Vehicle already registered 2: Vehicle not found in database 3: Terminal already registered 4: Terminal not found in database

3	Authentication code	STRING	Present only when successful
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2013 Version Example:7E **81 00** 00 0B 01 38 12 34 56 78 00 01 00 01 00 31 32 33 34 35 36 37 38 B3 7E

2019 Version Example:7E **81 00** 40 0B 01 00 00 00 00 01 38 12 34 56 78 00 01 00 01 00 31 32 33 34 35 36 37 38 F2 7E

0x0102 Terminal Authentication (Uplink)

[2011/2013]:

Offset	Field	Type	Description
0	Authentication code	STRING	Reported after terminal reconnection

[2019] Adds fields:

Offset	Field	Type	Description
0	Authentication code length	BYTE	
1	Authentication code content	STRING	
1+n	Terminal IMEI	BYTE[15]	[2019] Added
16+n	Software version number	BYTE[20]	[2019] Added

2011/2013 Version Example:7E **01 02** 00 09 01 38 12 34 56 78 00 01 08 31 32 33 34 35 36 37 38 3A 7E

2019 Version Example:7E **01 02** 40 2C 01 00 00 00 00 01 38 12 34 56 78 00 01 08 31 32 33 34 35 36 37 38 39 30 31 32 33 34 35 56 31 2E 30 30 2E 30 30 00 00 00 00 00 00 00 00 00 00 00 09 7E

0x8103 Set Terminal Parameters (Downlink)

Offset	Field	Type	Description
0	Total parameters	BYTE	
1	Parameter item list	—	See parameter item format

Parameter Item Format:

Field	Type	Description
Parameter ID	DWORD	See Appendix A for definitions
Parameter length	BYTE	Number of bytes of parameter value
Parameter value	—	Multiple parameter items with the same ID for multi-value parameters

Common parameter IDs are in Appendix A.

2013 Version Example: 7E 81 03 00 13 01 38 12 34 56 78 00 01 02 00 00 00 01 04 00 00 00 3C 00 00 00 20 04 00 00 00 00 BE 7E

2019 Version Example: 7E 81 03 40 13 01 00 00 00 00 01 38 12 34 56 78 00 01 02 00 00 00 01 04 00 00 00 3C 00 00 00 20 04 00 00 00 00 FF 7E

0x8104 Query Terminal Parameters (Downlink)

Message body is empty. Terminal responds with 0x0104.

2013 Version Example: 7E 81 04 00 00 01 38 12 34 56 78 00 01 B5 7E

2019 Version Example: 7E 81 04 40 00 01 00 00 00 00 01 38 12 34 56 78 00 01 F4 7E

0x8106 Query Specified Terminal Parameters (Downlink) [2019]

Offset	Field	Type	Description
0	Total parameters	BYTE	

1	Parameter ID list	DWORD[n]	Parameter IDs in order
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Terminal responds with 0x0104.

2013 Version Example:

(N/A, this message does not exist in 2013 version)

2019 Version Example: 7E **81 06** 40 09 01 00 00 00 00 01 38 12 34 56 78 00 01 02 00 00 00 01 00 00 00 20 DC 7E

0x0104 Query Terminal Parameters Response (Uplink)

Offset	Field	Type	Description
0	Response serial number	WORD	Corresponds to the query message serial number
2	Number of response parameters	BYTE	
3	Parameter item list	—	Format same as parameter item in Set Terminal Parameters

2013 Version Example: 7E **01 04** 00 0C 01 38 12 34 56 78 00 01 00 01 01 00 00 00 01 04 00 00 00 3C 00 7E

2019 Version Example: 7E **01 04** 40 0C 01 00 00 00 00 01 38 12 34 56 78 00 01 00 01 01 00 00 00 01 04 00 00 00 3C 41 7E

0x8105 Terminal Control (Downlink)

Offset	Field	Type	Description
0	Command word	BYTE	See command word table

1	Command parameters	STRING	Fields separated by ",," ;
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Command Word:

Value	2013	2019
1	Wireless upgrade	Wireless upgrade
2	Connect to specified server	Connect to specified server
3	Terminal power off	[2019] Deleted
4	Terminal reset	Terminal reset
5	Restore factory settings	Restore factory settings
6	Close data communication	[2019] Deleted
7	Close all wireless communication	[2019] Deleted

2013 Version Example:7E **81 05** 00 01 01 38 12 34 56 78 00 01 04 B1 7E

2019 Version Example:7E **81 05** 40 01 01 00 00 00 00 01 38 12 34 56 78 00 01 04 F0 7E

0x8107 Query Terminal Attributes (Downlink)

Message body is empty. Terminal responds with 0x0107.

2013 Version Example:7E **81 07** 00 00 01 38 12 34 56 78 00 01 B6 7E

2019 Version Example:7E **81 07** 40 00 01 00 00 00 00 01 38 12 34 56 78 00 01 F7 7E

0x0107 Query Terminal Attributes Response (Uplink)

Offset	Field	Type	Description
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0	Terminal type	WORD	bit0: Passenger transport bit1: Hazardous materials bit2: General freight bit3: Taxi bit6: Hard disk video recorder bit7: All-in-one machine
2	Manufacturer ID	BYTE[5]	
7	Terminal model	BYTE[30]	Pad with 0x00 if insufficient
37	Terminal ID	BYTE[30]	Pad with 0x00 if insufficient
67	Terminal SIM card ICCID	BCD[10]	
77	Terminal hardware version length	BYTE	n
78	Terminal hardware version	STRING	
78+n	Terminal firmware version length	BYTE	m
79+n	Terminal firmware version	STRING	
79+n+m	GNSS module attributes	BYTE	bit0: GPS bit1: BeiDou bit2: GLONASS bit3: Galileo

0x0200 Location Information Reporting (Uplink)

Message body = Location basic information (28 bytes) + Location additional information item list (optional).

Location Basic Information:

Offset	Field	Type	Description
0	Alarm flag	DWORD	See alarm flag bit definition below
4	Status	DWORD	See status bit definition below
8	Latitude	DWORD	Degrees × 10 ⁶
12	Longitude	DWORD	Degrees × 10 ⁶
16	Altitude	WORD	Altitude in meters
18	Speed	WORD	1/10 km/h
20	Direction	WORD	0~359, north is 0, clockwise
21	Time	BCD[6]	YY-MM-DD-hh-mm-ss, GMT+8

2013 Version Example:7E 02 00 00 1C 01 38 12 34 56 78 00 01 00 00 00 00 00 00 00 02 61 2D 9C 06 EF FD 20 00 32 00 96 00 5A 26 04 22 10 30 00 16 7E

2019 Version Example:7E 02 00 40 1C 01 00 00 00 00 01 38 12 34 56 78 00 01 00 00 00 00 00 00 02 61 2D 9C 06 EF FD 20 00 32 00 96 00 5A 26 04 22 10 30 00 57 7E

Alarm Flag Bit Definition (DWORD 32 bits):

Bit	2013 Definition	2019 Change
0	Emergency alarm	Same
1	Overspeed alarm	Same
2	Fatigue driving	Same

3	Danger warning	[2019] Changed to dangerous driving behavior alarm
4	GNSS module fault	Same
5	GNSS antenna not connected/cut	Same
6	GNSS antenna short circuit	Same
7	Terminal main power undervoltage	Same
8	Terminal main power outage	Same
9	LCD/Display fault	Same
10	TTS module fault	Same
11	Camera fault	Same
12	IC card module fault	Same
13	Overspeed warning	Same
14	Fatigue driving warning	Same
15	—	[2019] Illegal driving alarm
16	—	[2019] Tire pressure warning
17	—	[2019] Right turn blind spot anomaly alarm
18	Daily cumulative driving timeout	Same
19	Overtime parking	Same
20	In/out area	Same
21	In/out route	Same

22	Road segment driving time insufficient/too long	Same
23	Route deviation	Same
24	Vehicle VSS fault	Same
25	Vehicle fuel anomaly	Same
26	Vehicle stolen	Same
27	Vehicle illegal ignition	Same
28	Vehicle illegal displacement	Same
29	Collision warning	[2019] Changed to collision rollover alarm
30	Rollover warning	Same
31	Illegal door opening alarm	[2019] Reserved

Processing rules: Clear after response received (bits 0, 3, 20~22, 27~28); maintain until condition resolved (other bits).

Status Bit Definition (DWORD 32 bits):

Bit	Definition
0	ACC (0: Off; 1: On)
1	Positioning (0: Not positioned; 1: Positioned)
2	Latitude (0: North; 1: South)
3	Longitude (0: East; 1: West)
4	Operating status (0: Operating; 1: Out of service)
5	Latitude/longitude encrypted (0: Not encrypted; 1: Encrypted)
6	[2019] Forward collision warning system pre-collision alarm

7	[2019] Lane departure warning
8~9	Load (00: Empty; 01: Half load; 10: Reserved; 11: Full load)
10	Fuel line (0: Normal; 1: Disconnected)
11	Circuit (0: Normal; 1: Disconnected)
12	Door lock (0: Unlock; 1: Lock)
13	Door 1/Front door (0: Closed; 1: Open)
14	Door 2/Middle door (0: Closed; 1: Open)
15	Door 3/Rear door (0: Closed; 1: Open)
16	Door 4/Driver seat (0: Closed; 1: Open)
17	Door 5/Custom (0: Closed; 1: Open)
18	GPS positioning
19	BeiDou positioning
20	GLONASS positioning
21	Galileo positioning
22	[2019] Driving status (0: Stopped; 1: Driving)

Location Additional Information Item Format (TLV structure, can repeat multiple times, determined by message body length in message header):

Field	Type	Description
Additional information ID	BYTE	1~255
Additional information length	BYTE	Number of subsequent content bytes
Additional information	—	See additional information item definition below

Additional Information Item Definition:

ID	Length	Description
0x01	4	Mileage (DWORD, 1/10 km)
0x02	2	Fuel level (WORD, 1/10 L)
0x03	2	Driving record speed (WORD, 1/10 km/h)
0x04	2	Alarm event ID requiring manual confirmation (WORD)
0x05	30	[2019] Tire pressure (Pa)
0x06	2	[2019] Compartment temperature (°C)
0x11	1/5	Overspeed alarm additional info (location type BYTE + optional area ID DWORD)
0x12	6	In/out area/route alarm additional info (location type + area/route ID + direction BYTE)
0x13	7	Road segment driving time alarm additional info (road segment ID + driving time WORD + result BYTE)
0x25	4	Extended vehicle signal status bits, bit definition see table below
0x2A	2	IO status bits, bit definition see table below
0x2B	4	Analog quantity (AD0 + AD1)
0x30	1	Wireless communication network signal strength

0x31	1	GNSS positioning satellite count
0xE0	Variable	Custom information
0xE1~0xFF	—	Manufacturer defined

0x25 Extended Vehicle Signal Status Bits (DWORD 32 bits):

Bit	Definition
0	Low beam signal
1	High beam signal
2	Right turn signal
3	Left turn signal
4	Brake signal
5	Reverse signal
6	Fog light signal
7	Clearance lamp
8	Horn signal
9	Air conditioning status
10	Neutral gear signal
11	Retarder working
12	ABS working
13	Heater working
14	Clutch status
15~31	Reserved

0x2A IO Status Bits (WORD 16 bits):

Bit	Definition
0	Deep sleep status
1	Sleep status
2~15	Reserved

0x8201 Location Information Query (Downlink)

Message body is empty. Terminal responds with 0x0201.

2013 Version Example:7E 82 01 00 00 01 38 12 34 56 78 00 01 B3 7E

2019 Version Example:7E 82 01 40 00 01 00 00 00 00 01 38 12 34 56 78 00 01 F2 7E

0x0201 Location Information Query Response (Uplink)

Offset	Field	Type	Description
0	Response serial number	WORD	Corresponds to the query message serial number
2	Location report message body	—	Same as 0x0200 Location information reporting

2013 Version Example:7E 02 01 00 1E 01 38 12 34 56 78 00 01 00 01 00 00 00 00 00 00 00 02 61 2D 9C 06 EF FD 20 00 32 00 96 00 5A 26 04 22 10 30 00 14 7E

2019 Version Example:7E 02 01 40 1E 01 00 00 00 00 01 38 12 34 56 78 00 01 00 01 00 00 00 00 00 00 02 61 2D 9C 06 EF FD 20 00 32 00 96 00 5A 26 04 22 10 30 00 55 7E

0x0704 Batch Location Data Upload (Uplink)

Offset	Field	Type	Description
0	Number of data items	WORD	

2	Location data type	BYTE	0: Normal batch location report 1: Blind area supplemental report
3	Location report data item	—	See location report data item format

Location Report Data Item Format:

Offset	Field	Type	Description
0	Location report data body length	WORD	
2	Location report data body	BYTE[n]	See 0x0200 location information reporting, n is the location report data body length

2013 Version Example: 7E 07 04 00 21 01 38 12 34 56 78 00 01 00 01 00 00 1C 00 00 00 00 00 00 00 02 61 2D 9C 06 EF FD 20 00 32 00 96 00 5A 26 04 22 10 30 00 37 7E

2019 Version Example: 7E 07 04 40 21 01 00 00 00 00 01 38 12 34 56 78 00 01 00 01 00 00 1C 00 00 00 00 00 00 00 02 61 2D 9C 06 EF FD 20 00 32 00 96 00 5A 26 04 22 10 30 00 76 7E

0x8202 Temporary Location Tracking Control (Downlink)

Offset	Field	Type	Description
0	Time interval	WORD	Seconds, 0 stops tracking
2	Valid period	DWORD	Seconds

2013 Version Example: 7E 82 02 00 06 01 38 12 34 56 78 00 01 00 1E 00 00 01 2C 85 7E

2019 Version Example: 7E 82 02 40 06 01 00 00 00 00 01 38 12 34 56 78 00 01 00 1E 00 00 01 2C C4 7E

0x8203 Manual Alarm Confirmation (Downlink) [2019]

Offset	Field	Type	Description
0	Alarm message serial number	WORD	0 means all messages of this type
2	Manual confirmation alarm type	DWORD	See table below

Manual Confirmation Alarm Type Bit Definition (DWORD 32 bits):

Bit	Definition
0	Emergency alarm
1	Overspeed alarm
2	Fatigue driving
3	Dangerous driving behavior alarm
4	GNSS module fault
5	GNSS antenna not connected/cut
6	GNSS antenna short circuit
7	Terminal main power undervoltage
8	Terminal main power outage
9	LCD/Display fault
10	TTS module fault
11	Camera fault
12	IC card module fault
13	Overspeed warning
14	Fatigue driving warning

15	Illegal driving alarm
16	Tire pressure warning
17	Right turn blind spot anomaly alarm
18	Daily cumulative driving timeout
19	Overtime parking
20	In/out area
21	In/out route
22	Road segment driving time insufficient/too long
23	Route deviation
24	Vehicle VSS fault
25	Vehicle fuel anomaly
26	Vehicle stolen
27	Vehicle illegal ignition
28	Vehicle illegal displacement
29	Collision rollover alarm
30	Rollover warning
31	Illegal door opening alarm

2013 Version Example:

(N/A, this message does not exist in 2013 version)

2019 Version Example: 7E 82 03 40 06 01 00 00 00 00 01 38 12 34 56 78 00 01 00 01 00 00 00 01 F6 7E

0x8204 Link Detection (Downlink) [2019]

Message body is empty. Terminal responds with general response after receiving, used to detect link connectivity.

2013 Version Example:

(N/A, this message does not exist in 2013 version)

2019 Version Example:7E 82 04 40 00 01 00 00 00 00 01 38 12 34 56 78 00 01 F7 7E

4.3 Information

0x8300 Text Information下发 (Downlink)

2013:

Offset	Field	Type	Description
0	Flag	BYTE	bit0: Urgent bit2: Display on screen bit3: TTS playback bit4: Ad screen bit5: Navigation/CAN fault code
1	Text information	STRING	Max 1024 bytes, GBK

[2019] Adds text type field:

Offset	Field	Type	Description
0	Flag	BYTE	Same as above
1	Text type	BYTE	1: Notice 2: Service
2	Text information	STRING	Max 1024 bytes, GBK

2013 Version Example:7E 83 00 00 0A 01 38 12 34 56 78 00 01 01 01 B2 E2 CA D4 CF
FB CF A2 AE 7E

2019 Version Example:7E 83 00 40 0A 01 00 00 00 00 01 38 12 34 56 78 00 01 01 01 B2
E2 CA D4 CF FB CF A2 EF 7E

4.4 Telephone

0x8400 Telephone Callback (Downlink)

Offset	Field	Type	Description
0	Flag	BYTE	0: Normal call 1: Monitoring
1	Phone number	STRING	Max 20 bytes

2013 Version Example:7E 84 00 00 19 01 38 12 34 56 78 00 01 00 31 33 38 30 30 31 33 38 30 30 00 31 33 38 30 30 31 33 38 30 30 30 00 AD 7E

2019 Version Example:7E 84 00 40 19 01 00 00 00 00 01 38 12 34 56 78 00 01 00 31 33 38 30 30 31 33 38 30 30 31 33 38 30 30 30 00 EC 7E

0x8401 Set Telephone Book (Downlink)

Offset	Field	Type	Description
0	Set type	BYTE	0: Delete all 1: Update 2: Append 3: Modify
1	Total contacts	BYTE	
2	Contact items	—	See contact format

Contact Format:

Field	Type	Description
Flag	BYTE	1: Incoming 2: Outgoing 3: Both incoming and outgoing

Phone number length	BYTE	
Phone number	STRING	
Name length	BYTE	
Contact name	STRING	

2013 Version Example:7E 84 01 00 14 01 38 12 34 56 78 00 01 01 01 01 0B 31 33 38 30 30 31 33 38 30 30 30 04 D5 C5 C8 FD BA 7E

2019 Version Example:7E 84 01 40 14 01 00 00 00 00 01 38 12 34 56 78 00 01 01 01 01 0B 31 33 38 30 30 31 33 38 30 30 30 04 D5 C5 C8 FD FB 7E

4.5 Vehicle Control

0x8500 Vehicle Control (Downlink)

2013:

Offset	Field	Type	Description
0	Control flag	BYTE	bit0: 0 Door unlock/1 Door lock

[2019]:

Offset	Field	Type	Description
0	Number of control types	WORD	
2	Control type list	—	Each control type contains control type ID (WORD) + control parameters

Control Type ID Definition: 0x0001: Door
0xF001~0xFFFF: Manufacturer defined

Control Parameters (when control type ID is 0x0001): BYTE, 0: Door lock closed 1: Door lock open

2013 Version Example:7E 85 00 00 01 01 38 12 34 56 78 00 01 01 B5 7E

2019 Version Example:7E 85 00 40 05 01 00 00 00 00 01 38 12 34 56 78 00 01 00 01 00 01 01 F0 7E

0x0500 Vehicle Control Response (Uplink)

Offset	Field	Type	Description
0	Response serial number	WORD	Corresponds to the control message serial number
2	Location report message body	—	Same as 0x0200

2013 Version Example:7E 05 00 00 1E 01 38 12 34 56 78 00 01 00 01 00 00 00 00 00 00 00 02 61 2D 9C 06 EF FD 20 00 32 00 96 00 5A 26 04 22 10 30 00 12 7E

2019 Version Example:7E 05 00 40 1E 01 00 00 00 00 01 38 12 34 56 78 00 01 00 01 00 00 00 00 00 00 02 61 2D 9C 06 EF FD 20 00 32 00 96 00 5A 26 04 22 10 30 00 53 7E

4.6 Vehicle Management (Areas/Routes)

0x8600 Set Circular Area (Downlink)

Offset	Field	Type	Description
0	Set attribute	BYTE	0: Update 1: Append 2: Modify
1	Total areas	BYTE	
2	Area item list	—	See circular area item

Circular Area Item:

Field	Type	Description
Area ID	DWORD	
Area attribute	WORD	See area attribute bit definition
Center latitude	DWORD	Degrees × 10 ⁶
Center longitude	DWORD	Degrees × 10 ⁶
Radius	DWORD	Meters
Start time	BCD[6]	Optional
End time	BCD[6]	Optional
Maximum speed	WORD	Optional
Overspeed duration	BYTE	Optional, seconds
Night maximum speed	WORD	[2019] Optional, if area attribute bit1 is 0 then this field does not exist
Name length	WORD	[2019] Optional
Area name	STRING	[2019] Optional

Area Attribute Bits (WORD):

Bit	Definition
0	Time-based
1	Speed limit
2	Alarm driver when entering area
3	Alarm platform when entering area
4	Alarm driver when exiting area
5	Alarm platform when exiting area

8	Door opening prohibited
14	Close communication when entering area
15	Collect GNSS when entering area

[2019] Circular area item adds name field.

2013 Version Example: 7E **86 00** 00 14 01 38 12 34 56 78 00 01 00 01 00 00 00 01 00 01 02 60 D3 60 06 F0 1F 80 00 00 03 E8 F0 7E

2019 Version Example: 7E **86 00** 40 14 01 00 00 00 00 01 38 12 34 56 78 00 01 00 01 00 00 00 01 00 01 02 60 D3 60 06 F0 1F 80 00 00 03 E8 B1 7E

0x8601 Delete Circular Area (Downlink)

Offset	Field	Type	Description
0	Number of areas	BYTE	≤125, 0 means delete all
1	Area ID list	DWORD[n]	

2013 Version Example: 7E **86 01** 00 05 01 38 12 34 56 78 00 01 01 00 00 00 01 B2 7E

2019 Version Example: 7E **86 01** 40 05 01 00 00 00 00 01 38 12 34 56 78 00 01 01 00 00 00 01 F3 7E

0x8602 Set Rectangular Area (Downlink)

Offset	Field	Type	Description
0	Set attribute	BYTE	Same as circular area
1	Total areas	BYTE	
2	Area item list	—	See rectangular area item

Rectangular Area Item:

Field	Type	Description
-------	------	-------------

Area ID	DWORD	
Area attribute	WORD	See area attribute bit definition
Upper left latitude	DWORD	Degrees × 10 ⁶
Upper left longitude	DWORD	Degrees × 10 ⁶
Lower right latitude	DWORD	Degrees × 10 ⁶
Lower right longitude	DWORD	Degrees × 10 ⁶
Start time	BCD[6]	Optional
End time	BCD[6]	Optional
Maximum speed	WORD	Optional
Overspeed duration	BYTE	Optional, seconds

[2019] Rectangular area item adds name field.

2013 Version Example:7E **86 02** 00 18 01 38 12 34 56 78 00 01 00 01 00 00 00 01 00 01 02 60 D3 60 06 EE 98 E0 02 5F 4C C0 06 F1 A6 20 4C 7E

2019 Version Example:7E **86 02** 40 18 01 00 00 00 00 01 38 12 34 56 78 00 01 00 01 00 00 00 01 00 01 02 60 D3 60 06 EE 98 E0 02 5F 4C C0 06 F1 A6 20 0D 7E

0x8603 Delete Rectangular Area (Downlink)

Offset	Field	Type	Description
0	Number of areas	BYTE	≤125, 0 means delete all
1	Area ID list	DWORD[n]	

2013 Version Example:7E **86 03** 00 05 01 38 12 34 56 78 00 01 01 00 00 00 01 B0 7E

2019 Version Example:7E **86 03** 40 05 01 00 00 00 00 01 38 12 34 56 78 00 01 01 00 00 00 01 F1 7E

0x8604 Set Polygon Area (Downlink)

Offset	Field	Type	Description
0	Area ID	DWORD	
4	Area attribute	WORD	Same as circular area
6	Start time	BCD[6]	Optional
12	End time	BCD[6]	Optional
18	Maximum speed	WORD	Optional
20	Overspeed duration	BYTE	Optional
21	Total vertex count	WORD	
23+n	Night maximum speed	WORD	[2019] Optional, n is the total vertex length
25+n	Vertex items	—	Latitude (DWORD) + Longitude (DWORD)

[2019] Adds: Area name length (BYTE) + Area name (STRING)

2013 Version Example: 7E **86 04** 00 10 01 38 12 34 56 78 00 01 00 00 00 01 00 01 00 01 02 60 D3 60 06 EE 98 E0 E2 7E

2019 Version Example: 7E **86 04** 40 10 01 00 00 00 00 01 38 12 34 56 78 00 01 00 00 00 01 00 01 00 01 02 60 D3 60 06 EE 98 E0 A3 7E

0x8605 Delete Polygon Area (Downlink)

Offset	Field	Type	Description
0	Number of areas	BYTE	≤125, 0 means delete all
1	Area ID list	DWORD[n]	

2013 Version Example: 7E **86 05** 00 05 01 38 12 34 56 78 00 01 01 00 00 00 01 B6 7E

2019 Version Example: 7E 86 05 40 05 01 00 00 00 00 01 38 12 34 56 78 00 01 01 00 00
00 01 F7 7E

0x8606 Set Route (Downlink)

Offset	Field	Type	Description
0	Route ID	DWORD	
4	Route attribute	WORD	bit0: Time-based bit2: Alarm driver when entering route bit3: Alarm platform when entering route bit4: Alarm driver when exiting route bit5: Alarm platform when exiting route
6	Start time	BCD[6]	Optional
12	End time	BCD[6]	Optional
18	Total waypoint count	WORD	
20	Waypoint items	—	See waypoint item format

Waypoint Item:

Field	Type	Description
Waypoint ID	DWORD	
Road segment ID	DWORD	
Latitude	DWORD	Degrees × 10 ⁶
Longitude	DWORD	Degrees × 10 ⁶

Road segment width	BYTE	Meters
Road segment attribute	BYTE	
Driving too long threshold	WORD	Optional, seconds
Driving too short threshold	WORD	Optional, seconds
Maximum speed	WORD	Optional
Overspeed duration	BYTE	Optional, seconds

[2019] Adds: Route name length (BYTE) + Route name (STRING)

2013 Version Example: 7E 86 06 00 1A 01 38 12 34 56 78 00 01 00 00 00 01 00 01 00 01 00 00 00 01 00 00 00 01 02 60 D3 60 06 EE 98 E0 14 00 FE 7E

2019 Version Example: 7E 86 06 40 1A 01 00 00 00 00 01 38 12 34 56 78 00 01 00 00 00 01 00 01 00 01 00 00 00 01 00 00 00 01 02 60 D3 60 06 EE 98 E0 14 00 BF 7E

0x8607 Delete Route (Downlink)

Offset	Field	Type	Description
0	Number of routes	BYTE	≤125, 0 means delete all
1	Route ID list	DWORD[n]	

2013 Version Example: 7E 86 07 00 05 01 38 12 34 56 78 00 01 01 00 00 00 01 B4 7E

2019 Version Example: 7E 86 07 40 05 01 00 00 00 00 01 38 12 34 56 78 00 01 01 00 00 00 01 F5 7E

0x8608 Query Area or Route Data (Downlink) [2019]

Offset	Field	Type	Description
0	Query type	BYTE	1=Query circular area data, 2=Query rectangular area data, 3=Query polygon area data, 4=Query route data

1	Number of IDs to query	DWORD	0 means query all area or route data, greater than 0 means followed by IDs to query
5	ID1	DWORD	
9	ID2	DWORD	
$1+(n-1)\times 4$	IDn	DWORD	n is the number of IDs

2013 Version Example:

(N/A, this message does not exist in 2013 version)

2019 Version Example: 7E 86 08 40 05 01 00 00 00 00 01 38 12 34 56 78 00 01 01 00 00 00 00 FB 7E

0x0608 Query Area or Route Data Response (Uplink) [2019]

Offset	Field	Type	Description
0	Query type	BYTE	Same as 0x8608 query type
1	Number of returned data items	DWORD	Represents the number of returned area or route data items
5	Data item list	—	Each data item format is determined by the query type

Data Item Format:

Query Type	Data Structure
1	Circular area data structure (see 0x8600)

2	Rectangular area data structure (see 0x8602)
3	Polygon area data structure (see 0x8604)
4	Route data structure (see 0x8606)

2013 Version Example:

(N/A, this message does not exist in 2013 version)

2019 Version Example: 7E 06 08 40 05 01 00 00 00 00 01 38 12 34 56 78 00 01 01 00 00 00 00 7B 7E

4.7 Multimedia

0x0800 Multimedia Event Information Upload (Uplink)

Offset	Field	Type	Description
0	Multimedia ID	DWORD	>0
4	Multimedia type	BYTE	0: Image 1: Audio 2: Video
5	Format encoding	BYTE	0: JPEG 1: TIF 2: MP3 3: WAV 4: WMV

6	Event code	BYTE	0: Platform command 1: Timed 2: Robbery alarm 3: Collision rollover 4: Door open 5: Door close 6: Vehicle speed change 7: Fixed distance
7	Channel ID	BYTE	

2013 Version Example:7E 08 00 00 08 01 38 12 34 56 78 00 01 00 00 00 01 00 00 01 01 31 7E

2019 Version Example:7E 08 00 40 08 01 00 00 00 00 01 38 12 34 56 78 00 01 00 00 00 01 00 00 01 01 70 7E

0x0801 Multimedia Data Upload (Uplink)

Offset	Field	Type	Description
0	Multimedia ID	DWORD	
4	Multimedia type	BYTE	Same as above
5	Format encoding	BYTE	Same as above
6	Event code	BYTE	Same as above
7	Channel ID	BYTE	
8	Location information	BYTE[28]	Location basic information
36	Multimedia data	—	

2013 Version Example:7E 08 01 00 2E 01 38 12 34 56 78 00 01 00 00 00 01 00 00 01 01 00 00 00 00 00 00 00 00 02 61 2D 9C 06 EF FD 20 00 32 00 96 00 5A 26 04 22 10 30 00 00 00 00 00 00 00 00 00 00 00 00 2E 7E

2019 Version Example: 7E 08 01 40 2E 01 00 00 00 00 01 38 12 34 56 78 00 01 00 00 00 01 00 00 01 01 00 00 00 00 00 00 00 02 61 2D 9C 06 EF FD 20 00 32 00 96 00 5A 26 04 22 10 30 00 00 00 00 00 00 00 00 00 00 00 00 6F 7E

0x8800 Multimedia Data Upload Response (Downlink)

Offset	Field	Type	Description
0	Multimedia ID	DWORD	If all received, no subsequent fields
4	Total resend packages	BYTE	
5	Resend package ID list	BYTE[2*n]	

2013 Version Example: 7E 88 00 00 05 01 38 12 34 56 78 00 01 00 00 00 01 00 BC 7E

2019 Version Example: 7E 88 00 40 05 01 00 00 00 00 01 38 12 34 56 78 00 01 00 00 00 01 00 FD 7E

0x8801 Camera Immediate Capture Command (Downlink)

Offset	Field	Type	Description
0	Channel ID	BYTE	>0
1	Capture command	WORD	0: Stop 0xFFFF: Video recording Other: Number of photos
3	Interval/time	WORD	Seconds, 0 is minimum interval/continuous recording
5	Save flag	BYTE	1: Save 0: Real-time upload

6	Resolution	BYTE	0x00: Lowest resolution 0x01: 320×240 0x02: 640×480 0x03: 800×600 0x04: 1024×768 0x05: 176×144 0x06: 352×288 0x07: 704×288 0x08: 704×576
7	Image quality	BYTE	1~10, 1 is best
8	Brightness	BYTE	0~255
9	Contrast	BYTE	0~127
10	Saturation	BYTE	0~127
11	Chroma	BYTE	0~255

2013 Version Example:7E **88 01** 00 0B 01 38 12 34 56 78 00 01 01 00 01 00 05 01 02 05 64 00 01 D4 7E

2019 Version Example:7E **88 01** 40 0B 01 00 00 00 00 01 38 12 34 56 78 00 01 01 00 01 00 05 01 02 05 64 00 01 95 7E

0x0805 Camera Immediate Capture Command Response (Uplink)

Offset	Field	Type	Description
0	Response serial number	WORD	Corresponds to the capture command serial number
2	Result	BYTE	0: Success 1: Failure 2: Channel not supported
3	Number of IDs	WORD	Valid when successful

5	ID list	DWORD[n]	Each multimedia ID is 4 bytes
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2013 Version Example:7E **08 05** 00 09 01 38 12 34 56 78 00 01 00 01 00 00 01 00 00 00 01 35 7E

2019 Version Example:7E **08 05** 40 09 01 00 00 00 00 01 38 12 34 56 78 00 01 00 01 00 00 01 00 00 00 01 74 7E

0x8802 Stored Multimedia Data Search (Downlink)

Offset	Field	Type	Description
0	Multimedia type	BYTE	0: Image 1: Audio 2: Video
1	Channel ID	BYTE	0: All channels
2	Event code	BYTE	
3	Start time	BCD[6]	
9	End time	BCD[6]	

2013 Version Example:7E **88 02** 00 0F 01 38 12 34 56 78 00 01 00 01 01 26 04 22 10 30 00 26 04 22 10 30 00 B5 7E

2019 Version Example:7E **88 02** 40 0F 01 00 00 00 00 01 38 12 34 56 78 00 01 00 01 01 26 04 22 10 30 00 26 04 22 10 30 00 F4 7E

0x0802 Stored Multimedia Data Search Response (Uplink)

Offset	Field	Type	Description
0	Response serial number	WORD	
2	Total items	WORD	
4	Search items	—	See multimedia search item format, can repeat

Multimedia Search Item Format:

Offset	Field	Type	Description
0	Multimedia ID	DWORD	>0
4	Multimedia type	BYTE	0: Image 1: Audio 2: Video
5	Channel ID	BYTE	
6	Event code	BYTE	0: Platform command 1: Timed 2: Robbery alarm 3: Collision rollover
7	Location information report message body	—	Same as 0x0200 Location basic information

2013 Version Example:7E 08 02 00 27 01 38 12 34 56 78 00 01 00 01 00 01 00 00 00 01 00 01 01 00 00 00 00 00 00 00 02 61 2D 9C 06 EF FD 20 00 32 00 96 00 5A 26 04 22 10 30 00 24 7E

2019 Version Example:7E 08 02 40 27 01 00 00 00 00 01 38 12 34 56 78 00 01 00 01 00 01 00 00 00 01 00 01 01 00 00 00 00 00 00 00 00 02 61 2D 9C 06 EF FD 20 00 32 00 96 00 5A 26 04 22 10 30 00 65 7E

0x8803 Stored Multimedia Data Upload Command (Downlink)

Offset	Field	Type	Description
0	Multimedia type	BYTE	
1	Channel ID	BYTE	
2	Event code	BYTE	
3	Start time	BCD[6]	
9	End time	BCD[6]	

15	Delete flag	BYTE	0: Retain 1: Delete
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2013 Version Example:7E **88 03** 00 10 01 38 12 34 56 78 00 01 00 01 01 26 04 22 10 30 00 26 04 22 10 30 00 00 AB 7E

2019 Version Example:7E **88 03** 40 10 01 00 00 00 00 01 38 12 34 56 78 00 01 00 01 01 26 04 22 10 30 00 26 04 22 10 30 00 00 EA 7E

0x8804 Recording Start Command (Downlink)

Offset	Field	Type	Description
0	Recording command	BYTE	0: Stop 1: Start
1	Recording time	WORD	Seconds, 0 means continuous recording
3	Save flag	BYTE	0: Real-time upload 1: Save
4	Sample rate	BYTE	0: 8K 1: 11K 2: 23K 3: 32K

2013 Version Example:7E **88 04** 00 05 01 38 12 34 56 78 00 01 01 00 3C 01 00 85 7E

2019 Version Example:7E **88 04** 40 05 01 00 00 00 00 01 38 12 34 56 78 00 01 01 00 3C 01 00 C4 7E

0x8805 Single Stored Multimedia Data Search Upload Command (Downlink)

Offset	Field	Type	Description
0	Multimedia ID	DWORD	>0

4	Delete flag	BYTE	0: Retain 1: Delete
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2013 Version Example:7E **88 05** 00 05 01 38 12 34 56 78 00 01 00 00 00 01 00 B9 7E

2019 Version Example:7E **88 05** 40 05 01 00 00 00 00 01 38 12 34 56 78 00 01 00 00 00 01 00 F8 7E

4.8 General Data Transmission

0x8900 Downlink Transparent Transmission (Downlink)

Offset	Field	Type	Description
0	Transparent message type	BYTE	See transparent type table
1	Transparent message content	—	

2013 Version Example:7E **89 00** 00 0B 01 38 12 34 56 78 00 01 00 00 00 00 00 00 00 00 00 00 B2 7E

2019 Version Example:7E **89 00** 40 0B 01 00 00 00 00 01 38 12 34 56 78 00 01 00 00 00 00 00 00 00 F3 7E

0x0900 Uplink Transparent Transmission (Uplink)

Format is the same as 0x8900.

Transparent Message Type:

Type Value	Description
0x00	GNSS module detailed positioning data
0x0B	Road transport certificate IC card information (upload 64B, download 24B, timeout 30s)
0x41	Serial port 1 transparent transmission

2019 Version Example:7E 8A 00 40 84 01 00 00 00 00 01 38 12 34 56 78 00 01 00 01 00
01 00
00
00
00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 7F 7E

Offset	Field	Type	Description
0	e	DWORD	RSA public key {e,n}'s e
4	n	BYTE[128]	RSA public key {e,n}'s n

2019 Version Example:7E 0A 00 40 84 01 00 00 00 00 01 38 12 34 56 78 00 01 00 01 00
01 00
00
00
00
00 00 00 00 00 00 00 00 00 00 00 00 00 FF 7E

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Corresponds to the parameter items in 0x8103 Set Terminal Parameters / 0x0104 Query Terminal Parameters Response. Parameter item format: Parameter ID (DWORD) + Parameter length (BYTE) + Parameter value.

Communication Timeout and Retransmission

Parameter ID	Type	Description
0x0001	DWORD	Terminal heartbeat sending interval (seconds)
0x0002	DWORD	TCP message response timeout (seconds)
0x0003	DWORD	TCP message retransmission count
0x0004	DWORD	UDP message response timeout (seconds)
0x0005	DWORD	UDP message retransmission count
0x0006	DWORD	SMS message response timeout (seconds)
0x0007	DWORD	SMS message retransmission count

Network Connection and Server

Parameter ID	Type	Description
0x0010	STRING	Primary server APN (for CDMA network, this is the PPP dial-up number)
0x0011	STRING	Primary server wireless communication dial username

0x0012	STRING	Primary server wireless communication dial password
0x0013	STRING	Primary server address (IP or domain), host port separated by colon, multiple servers separated by semicolon
0x0014	STRING	Backup server APN
0x0015	STRING	Backup server wireless communication dial username
0x0016	STRING	Backup server wireless communication dial password
0x0017	STRING	Backup server address, same format as 0x0013
0x0023	STRING	[2019] Secondary server APN, uses primary server configuration if empty
0x0024	STRING	[2019] Secondary server dial username
0x0025	STRING	[2019] Secondary server dial password
0x0026	STRING	[2019] Secondary server backup address

Server Port

Parameter ID	Type	Description
0x0018	DWORD	Server TCP port (reserved in 2019)

0x0019	DWORD	Server UDP port (reserved in 2019)
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IC Card Server

Parameter ID	Type	Description
0x001A	STRING	Road transport certificate IC card authentication primary server IP or domain
0x001B	DWORD	IC card authentication primary server TCP port
0x001C	DWORD	IC card authentication primary server UDP port
0x001D	STRING	IC card authentication backup server IP or domain, port same as primary server

Location Reporting Strategy

Parameter ID	Type	Description
0x0020	DWORD	Location reporting strategy: 0=Timed reporting, 1=Distance- based reporting, 2=Both timed and distance-based reporting
0x0021	DWORD	Location reporting scheme: 0=Based on ACC status, 1=Based on login status and ACC status
0x0022	DWORD	Driver not logged in reporting time interval (seconds)

0x0027	DWORD	Sleep mode reporting time interval (seconds)
0x0028	DWORD	Emergency alarm reporting time interval (seconds)
0x0029	DWORD	Default time reporting interval (seconds)
0x002C	DWORD	Default distance reporting interval (meters)
0x002D	DWORD	Driver not logged in reporting distance interval (meters)
0x002E	DWORD	Sleep mode reporting distance interval (meters)
0x002F	DWORD	Emergency alarm reporting distance interval (meters)

Waypoint and Fence

Parameter ID	Type	Description
0x0030	DWORD	Waypoint supplementary angle (<180)
0x0031	WORD	Electronic fence radius/Illegal displacement threshold (meters)
0x0032	BYTE[4]	[2019] Illegal driving time period (start/end time, 4 bytes)

Phone Numbers

Parameter ID	Type	Description
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0x0040	STRING	Monitoring platform phone number
0x0041	STRING	Reset phone number
0x0042	STRING	Factory reset phone number
0x0043	STRING	Monitoring platform SMS phone number
0x0044	STRING	Receive terminal SMS text alarm phone number
0x0045	DWORD	Phone answering strategy (0: Auto answer 1: Auto answer when ACC ON)
0x0046	DWORD	Maximum single call duration (seconds, 0 not allowed, 0xFFFFFFFF unlimited)
0x0047	DWORD	Maximum monthly call duration (seconds, 0 not allowed, 0xFFFFFFFF unlimited)
0x0048	STRING	Monitoring phone number
0x0049	STRING	Regulatory platform privilege SMS phone number

Alarm Control

Parameter ID	Type	Description
0x0050	DWORD	Alarm mask (corresponds to alarm flag bits, 1=masked)

0x0051	DWORD	Alarm send text SMS switch (corresponds to alarm flag bits, 1=send)
0x0052	DWORD	Alarm capture switch (corresponds to alarm flag bits, 1=capture)
0x0053	DWORD	Alarm capture storage flag (1=Store, 0=Real-time upload)
0x0054	DWORD	Key flag (corresponds to alarm flag bits, 1=Key alarm)
0x0055	DWORD	Maximum speed (km/h)
0x0056	DWORD	Overspeed duration (seconds)
0x0057	DWORD	Continuous driving time threshold (seconds)
0x0058	DWORD	Daily cumulative driving time threshold (seconds)
0x0059	DWORD	Minimum rest time (seconds)
0x005A	DWORD	Maximum parking time (seconds)
0x005B	WORD	Overspeed warning differential (1/10 km/h)
0x005C	WORD	Fatigue driving warning differential (seconds)
0x005D	WORD	Collision alarm parameters (b7~b0: Collision time ms b15~b8: Acceleration 0.1g, default 10)

0x005E	WORD	Rollover alarm angle (degrees, default 30°)
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Photo Control

Parameter ID	Type	Description
0x0064	DWORD	Timed photo control, bit definition see table below
0x0065	DWORD	Distance-based photo control, bit definition see table below

0x0064 Timed Photo Control Bit Definition (DWORD 32 bits):

Bit	Definition	Description
0	Channel 1 timed photo switch	0: Not allowed 1: Allowed
1	Channel 2 timed photo switch	0: Not allowed 1: Allowed
2	Channel 3 timed photo switch	0: Not allowed 1: Allowed
3	Channel 4 timed photo switch	0: Not allowed 1: Allowed
4	Channel 5 timed photo switch	0: Not allowed 1: Allowed
5~7	Reserved	
8	Channel 1 storage flag	0: Storage 1: Upload
9	Channel 2 storage flag	0: Storage 1: Upload

10	Channel 3 storage flag	0: Storage 1: Upload
11	Channel 4 storage flag	0: Storage 1: Upload
12	Channel 5 storage flag	0: Storage 1: Upload
13~15	Reserved	
16	Timed time unit	0: Seconds (<5s processed as 5s) 1: Minutes
17~31	Timed time interval	Executed after receiving parameter settings or restart

0x0065 Distance-based Photo Control Bit Definition (DWORD 32 bits):

Bit	Definition	Description
0	Channel 1 distance-based photo switch	0: Not allowed 1: Allowed
1	Channel 2 distance-based photo switch	0: Not allowed 1: Allowed
2	Channel 3 distance-based photo switch	0: Not allowed 1: Allowed
3	Channel 4 distance-based photo switch	0: Not allowed 1: Allowed
4	Channel 5 distance-based photo switch	0: Not allowed 1: Allowed
5~7	Reserved	
8	Channel 1 storage flag	0: Storage 1: Upload

9	Channel 2 storage flag	0: Storage 1: Upload
10	Channel 3 storage flag	0: Storage 1: Upload
11	Channel 4 storage flag	0: Storage 1: Upload
12	Channel 5 storage flag	0: Storage 1: Upload
13~15	Reserved	
16	Distance unit	0: Meters (<100m processed as 100m) 1: Kilometers
17~31	Distance interval	Executed after receiving parameter settings or restart

Image Quality

Parameter ID	Type	Description
0x0070	DWORD	Image/video quality (1~10, 1 is best)
0x0071	DWORD	Brightness (0~255)
0x0072	DWORD	Contrast (0~127)
0x0073	DWORD	Saturation (0~127)
0x0074	DWORD	Chroma (0~255)

Vehicle Information

Parameter ID	Type	Description
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0x0080	DWORD	Vehicle odometer reading (1/10 km)
0x0081	WORD	Vehicle province ID
0x0082	WORD	Vehicle city ID
0x0083	STRING	Motor vehicle license plate
0x0084	BYTE	License plate color (JT/T 697.7-2014, 0 if not registered)

GNSS Module

Parameter ID	Type	Description
0x0090	BYTE	GNSS positioning mode (bit0: GPS bit1: BeiDou bit2: GLONASS bit3: Galileo, 1=Enabled)
0x0091	BYTE	GNSS baud rate (0: 4800 1: 9600 2: 19200 3: 38400 4: 57600 5: 115200)
0x0092	BYTE	GNSS positioning data output frequency (0: 500ms 1: 1000ms default 2: 2000ms 3: 3000ms 4: 4000ms)
0x0093	DWORD	GNSS positioning data acquisition frequency (seconds, default 1)

0x0094	BYTE	GNSS data upload mode (0: Do not upload 1: By time 2: By distance 0x0B: Cumulative time 0x0C: Cumulative distance 0x0D: Cumulative count)
0x0095	DWORD	GNSS data upload settings (unit changes with upload mode: seconds/meters/count)

CAN Bus

Parameter ID	Type	Description
0x0100	DWORD	CAN channel 1 acquisition time interval (ms, 0 means no acquisition)
0x0101	WORD	CAN channel 1 upload time interval (seconds, 0 means do not upload)
0x0102	DWORD	CAN channel 2 acquisition time interval (ms, 0 means no acquisition)
0x0103	WORD	CAN channel 2 upload time interval (seconds, 0 means do not upload)
0x0110	BYTE[8]	CAN bus ID individual acquisition settings (bit63~32: Acquisition interval ms bit31: Channel number bit30: Frame type bit29: Acquisition mode bit28~0: CAN ID)

0x0111~0x01FF	BYTE[8]	Other CAN bus ID individual acquisition settings
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Custom

Parameter ID	Type	Description
0xF000~0xFFFF	—	Manufacturer defined